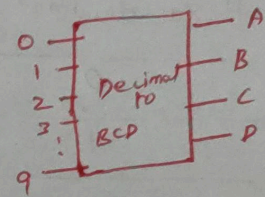


Encoder types

- i) priority encoder
- ii) Decimal to BCD encoder
- iii) Octal to Binary encoder
- iv) Hexadecimal to Binary encoder

$n=10$ (0 to 9)
Decimal to BCD encoder

i/p Decimal	output			
	D	C	B	A
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1

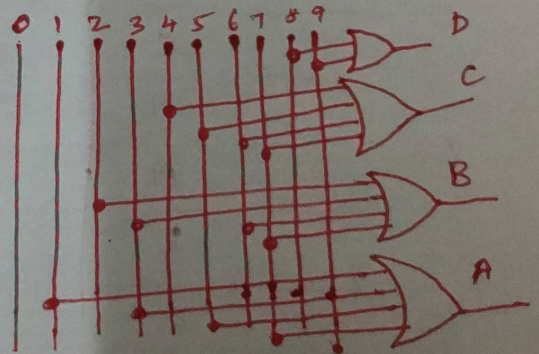


$$D = 8 + 9$$

$$C = 4 + 5 + 6 + 7$$

$$B = 2 + 3 + 6 + 7$$

$$A = 1 + 3 + 5 + 7 + 9$$



(0 to 7) (x y z)
Octal to Binary Encoder

$$x = D_4 + D_5 + D_6 + D_7$$

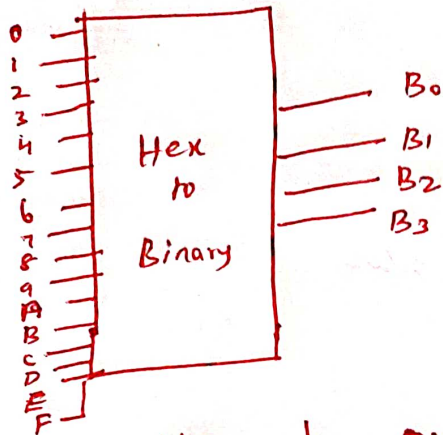
$$y = D_2 + D_3 + D_6 + D_7$$

$$z = D_1 + D_3 + D_5 + D_7$$

Hexadecimal to Binary encoder

$$8 = 16$$

0 to F
↓
15



input Hexa decimal	output Binary B3 B2 B1 B0
0	0 0 0 0
1	0 0 0 1
2	0 0 1 0
3	0 0 1 1
4	0 1 0 0
5	0 1 0 1
6	0 1 1 0
7	0 1 1 1
8	1 0 0 0
9	1 0 0 1
A	1 0 1 0
B	1 0 1 1
C	1 1 0 0
D	1 1 0 1
E	1 1 1 0
F	1 1 1 1

$$B_0 = 1 + 3 + 5 + 7 + 9 + B + D + F$$

$$B_1 = 2 + 3 + 6 + 7 + A + B + E + F$$

$$B_2 = 4 + 5 + 6 + 7 + C + D + E + F$$

$$B_3 = 8 + 9 + A + B + C + D + E + F$$

Priority Encoder

Truth table of a priority Encoder

Inputs				Outputs		
D ₀	D ₁	D ₂	D ₃	X	Y	V
0	0	0	0	X	X	0
1	0	0	0	0	0	1
X	1	0	0	0	1	1
X	X	1	0	1	0	1
X	X	X	1	1	1	1

V

0	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1

D₀D₁ D₂D₃

I/P				O/P		
D ₀	D ₁	D ₂	D ₃	X	Y	V
0	0	0	0	X	X	0
0	0	0	1	1	1	1
0	0	1	0	1	0	1
0	0	1	1	1	1	1
0	1	0	0	0	1	1
0	1	0	1	1	1	1
0	1	1	0	1	0	1
0	1	1	1	1	1	1
1	0	0	0	0	0	1
1	0	0	1	1	1	1
1	0	1	0	1	0	1
1	0	1	1	1	1	1
1	1	0	0	0	1	1
1	1	0	1	1	1	1
1	1	1	0	1	0	1
1	1	1	1	1	1	1

$D_3 \rightarrow 1 \Rightarrow 11$
 $D_2 \rightarrow 1 \rightarrow 10$
 $D_1 \rightarrow 1 \rightarrow 01$
 $D_0 \rightarrow 1 \rightarrow 00$

X

D ₀ D ₁	00	01	11	10
00	X	1	1	1
01	0	1	1	1
11	0	1	1	1
10	0	1	1	1

D₂D₃

$$X = D_3 + D_2$$

Y

D ₀ D ₁	00	01	11	10
00	X	1	1	0
01	1	1	1	0
11	1	1	1	0
10	0	1	1	1

D₂D₃

$$Y = D_3 + D_1 \overline{D_2}$$

Design a priority Encoder with 2, 1, 0, 3

D_2	D_1	D_0	D_3	X	Y
0	0	0	0	X	X
0	0	0	1	0	0
0	0	1	0	0	1
0	0	1	1	0	1
0	1	0	0	1	0
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	1	1	0
1	0	0	0	1	1
1	0	0	1	1	1
1	0	1	0	1	1
1	0	1	1	1	1
1	1	0	0	1	1
1	1	0	1	1	1
1	1	1	0	1	1
1	1	1	1	1	1

$D_3 \rightarrow 00$

$D_0 \rightarrow 01$

$D_1 \rightarrow 10$

$D_2 \rightarrow 11$

For X

$D_2 D_1$	$D_0 D_3$ 00	01	11	10
00	X	0	0	0
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$$X = D_1 + D_2$$

For Y

$D_2 D_1$	$D_0 D_3$ 00	01	11	10
00	X	0	1	1
01	0	0	0	0
11	1	1	1	1
10	1	1	1	1

$$Y = D_2 + \overline{D_1} D_0$$